

Modeling and Calibrating: LLMs Figure out their Decoding Process

Matthew Khoriaty

Reliable ML with Unreliable Black Boxes

June 4, 2026

The question

- An autoregressive LM is trained to *predict* text — so when it *generates*, it has an incentive to model its own **decoder** (temperature, top- k , ...).
- Two questions: **(a)** does it? **(b)** can we use that to predict better?

Can a model read the *temperature* of the text in front of it, and recalibrate itself?

The tool: a streaming Bayesian filter on $\log \beta$

- A Laplace filter maintains a posterior on $\eta = \log \beta$ (inverse-temperature), updated *per token* from the prefix — online, causal, training-free. $\hat{\beta}_t$ depends only on $x_{<t}$.
- Each token is a Newton / Fisher-scoring step on $P(x \mid \eta) = \text{softmax}(e^\eta \ell)[x]$.
- **Fix (this work):** the production filter linearises at $\beta = 1$ and is *inconsistent*. The *faithful* filter re-linearises at the running $\hat{\beta}$, with the correct β, β^2 chain-rule terms.

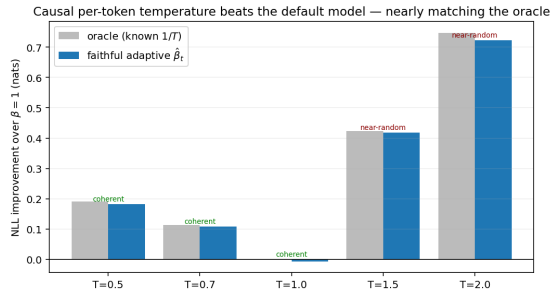
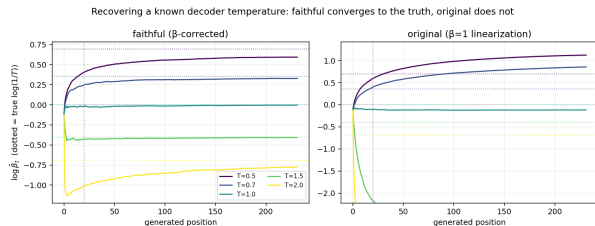
Two honest negatives (clearing the ground)

- **Closed-loop correction self-neutralises.** Feeding $\hat{\beta}$ back into the sampler drives $\beta_{\text{dec}} \rightarrow 1$ — the temperature is quietly thrown away. Confirmed across exact-grid filters (not an approximation artifact).
- **Real creative writing is already calibrated.** The best single temperature gives a non-significant fraction-of-a-nat gain — nothing to fix.

*So: to show self-calibration works, test where the temperature is **known** and there is something to recover.*

Claim 1: recovering a *known* decoder temperature

Generate Qwen-2.5-3B self-text at a known $\beta = 1/T$ (64 sequences per T); run the filter causally over it.



- **Recovery:** faithful $\hat{\beta}$ hits the truth (bias -0.030 at $T=0.7$); the original *diverges* (-4.150 at $T=2$).
- **Prediction:** the causal per-token temperature beats the default $\beta=1$ by 0.108 nats at $T=0.7$ (up to 0.723 at $T=2$), within $+0.025$ nats of the oracle.

But does natural text carry a temperature to recover?

text type	optimal T	$\Delta\text{NLL vs } \beta=1$ (nats)	$\beta=1$ NLL	$H_{\beta=1}$
shuffled WikiText	1.08	+0.0283**	7.20	6.48
Recamán (A005132)	0.91	+0.0032**	0.08	0.14
ABC music	1.03	+0.0019**	2.48	2.34
WritingPrompts	1.04	+0.0011	2.87	2.74
Telugu	1.03	+0.0010**	1.12	1.06
Esperanto	1.03	+0.0007	2.72	2.63
Yoruba	1.00	+0.0001	3.06	3.11
WikiText	1.00	-0.0000	1.96	1.93
Gutenberg (archaic EN)	1.01	-0.0001	1.87	1.85

- **Qwen is temperature-robust everywhere:** optimal $T \in [0.91, 1.08]$; gains $\leq +0.0032$ nats on real text (+0.0283 on shuffled gibberish).
- **Recamán (A005132)** is the exception: uniquely *underconfident* ($T = 0.91$) — right so often it should be *sharper*. “Harder $\neq$ mis-temperated” (Yoruba is hardest, yet $T \approx 1$).
- $\Rightarrow$ no meaningful miscalibration to exploit in the wild. (*Checked per-passage on the two worst types: a cross-validated per-passage temperature doesn't beat global, and the causal filter captures ≈ 0 .*)

Takeaways

- The *corrected* filter recovers a decoder temperature from context and predicts that text materially better than the default model — using only the model's own logits.
- Correct Bayesian bookkeeping (the β, β^2 chain-rule terms) is what makes it work; the original filter is inconsistent.
- Honest scope: an honest **positive** (Claim 1) *and* an honest **negative** — natural text is temperature-robust even far out of distribution.